Nanocourse Syllabus: Shiny Apps for Interactive Data Analysis & Sharing

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Dates: 9am – 5pm, August 4-5th 2025
Location: G9.250A

Summary:

The purpose of this course is to introduce students to the Shiny package for creating interactive applications with their biomedical research data. The nanocourse will be two complete days, with each morning filled with lectures and coding demonstrations. Most of each afternoon will be devoted to independent projects. Therefore, the goal for each student is to create a functional interactive app built with their own data and present that app to the class at the end of the last day. There are many details and advanced topics that are worth exploring, but this course will focus on teaching the basics of Shiny logic (server/UI connection), deployment to the web for effective sharing, and some commonly desired app features (interactive plots & file manipulation). We hope that this course will leave students with a tangible new computational skill and catalyze their independent learning of further Shiny techniques.

Schedule:

Day 1: Shiny Basics	
9 - 10 am	Intro / Shiny 101
10:15 – 11:15 am	User interface side
11:30 – 12:30 pm	Server side
1:30 – 2 pm	Case study
2 – 5 pm	Independent projects
Day 2: Shiny extras and deployment	
9 - 10:30 am	Interactive plots, files, extras
10:45 – 11:45 am	App deployment
12:45 – 3:45 pm	Independent projects
4 – 5 pm	Final presentations

Computing environment and prerequisites:

Please arrive at the nanocourse with a working laptop running R and Rstudio with the Tidyverse and Shiny packages installed. If you intend to also use Python, please have Visual Studio Code installed with the Shiny extension or a suitable alternative that can already produce a functional shiny app.

For your independent projects, please come with data in a usable format. We recommend that you have performed preliminary analysis on this data and your project is focused on creating an interactive app (not performing static analysis).

You should have a github account. We will use github to interact with course files and deploy certain shiny apps to the internet. If you want to make use of lab data on biohpc, you must have biohpc account and know how to navigate this resource.

The language of instruction will be R, so we expect that students will come into the class with at least a basic understanding of this programming language. Below are two relatively short books (free online) that may be useful for students to study before starting this course.

- Hands-On Programming with R
- R for Data Science

Shiny resources:

While we feel it is important for each student to setup a local computing environment, it is also worth pointing you towards excellent cloud environments that now exist. Some quick Shiny app development could take place entirely in the browser:

- Shinylive.io for R
- Shinylive.io for Python

The “textbook” for this course will be Mastering Shiny, which is an excellent resource, and we strongly recommend that students start here and use this to initially answer questions. The core Shiny websites developed by Posit are also very high-quality resources that we will use throughout the course and should be a default information source.

- Mastering Shiny Textbook (free online)
- Shiny homepage (Posit)